1.1.2 Real-world uses of generative AI

Let's look at some use cases to see exactly how this technology is so valuable in web development.

USE CASE: DIGITAL MARKETING AGENCY

Say a digital marketing agency wants to explore new creative possibilities for its client's advertising campaigns. The agency wants to create a toolbox that allows it to generate images from text, generate new images from existing images, and create virtual objects based on written prompts. Here's how the agency might use AI to achieve its goals:

- *Image generation*—The agency could integrate generative AI models like DALL-E into its web applications. By using external APIs from AI service providers, users can input text descriptions or prompts and generate corresponding images directly within the application.
- *Image-to-image translation*—The agency could use architectures like CycleGAN or Pix2Pix in its web applications for image-to-image translation. It also has options to use an external provider like stability.ai.
- *Text generation for marketing*—Utilizing state-of-the-art language models like GPT-4 or Llama, the agency intends to empower users to generate compelling marketing copy or product descriptions directly within the web application.

USE CASE: CUSTOMER EXPERIENCE MANAGEMENT

A customer support platform aims to enhance its customer experience management by integrating generative chatbots into its web application. Its AI solution includes

- *Generative chatbots for automated customer support*—The platform could integrate advanced chatbot models such as GPT into its web application to handle customer inquiries and provide real-time assistance.
- *Dynamic response generation for customer feedback*—To address customer feedback, the platform can rely on generative AI models for dynamic response generation using sentiment analysis APIs, such as Google Cloud Natural Language API or IBM Watson Natural Language Understanding, to analyze feedback sentiment. The platform can also use text generation models like GPT-4 via API integration to generate personalized responses based on the sentiment and context of the feedback.

USE CASE: MOCK INTERVIEW APPLICATION WITH GENERATIVE AI AGENTS

A job-training platform aims to enhance its mock interview feature using these tools:

- *AI interviewer agents for simulated interviews*—The platform will incorporate AI agents powered by generative AI models such as GPT-4 into its web application. By using external APIs for chatbot integration and natural language understanding, users can engage in simulated interview experiences in which AI interviewer agents ask questions, evaluate responses, and provide real-time feedback. Additionally, the platform integrates speech-to-text technology to enable users to speak their responses naturally, enhancing the authenticity of the interview experience.